AU-FUKAYA LAZY DEMOGRAPHIC ENGINE: A CATEGORY-THEORETIC ARCHITECTURE FOR REAL-TIME ADVERTISING PLACEMENT AT SCALE

Abstract. We present an advertising placement engine for large-scale transit networks grounded in the *classical (T=0) limit* of Fukaya-Oh-Ohta-Ono (FOOO) Floer theory over a toric symplectic manifold. The architecture implements three layers: (1) an AU attic of unmaterialised demographic coproducts (Joyal NNO); (2) a time-varying Hamiltonian flow; and (3) a pair-of-pants coproduct Δ evaluated on demand. The central claim is precise: our disk-count heuristic $\sqrt{L_i \cdot L_j} \cdot \omega$ is the *leading-order term* of the FOOO potential function $W(u)$ in the T-adic expansion. At this order $\partial^2 = 0$ holds vacuously, and the routing table has the structure of a constructible sheaf on the transit base in the sense of FLTZ. The gap to the full quantum theory — Novikov ring corrections, wall-crossing cluster mutations, Dehn twist monodromy — is identified precisely and constitutes the programme for Compiler Passes 6–8. We reduce memory from $O(|S||P|)$ (240 EB) to $O(|S| + |P|)$ ($\approx$400 GB), compile five passes producing SLA-compliant flat artifacts, and validate on the 81-station Hong Kong MTR network. The same classical-limit structure governs pharmacodynamic transport in the BALBc mouse brain connectome, establishing a unified framework.

Contents

Date: June 8, 2026.

1. Introduction

Modern advertising placement at scale faces a fundamental impossibility: any architecture that precomputes compatibility scores for all (station, product, demographic, time) combinations encounters a combinatorial wall at planetary scale. At 4×10^9 stations and 1.5×10^{10} products, a single-precision precomputed matrix requires

$$4 \times 10^9 \times 1.5 \times 10^{10} \times 4\,\text{bytes} = 240\,\text{exabytes of RAM},$$

exceeding total global storage capacity. Adding a 24-hour $\times$ 12-month time dimension inflates this to 67.5 zettabytes—not a scaling challenge but a proof of impossibility.

The architecture presented here sidesteps this wall entirely by treating advertising placement as a problem in *Fukaya-category-theoretic laziness*: demographic populations exist as formal coproduct symbols in a Joyal Arithmetic Universe and are never materialised unless an explicit algebraic operation at a surgery node demands them.

The Hong Kong MTR network serves as the validation domain. The pharmacodynamic correspondence (Section 11) connects the architecture to our earlier work on opioid transport in the BALBc mouse brain connectome, establishing that the same AU-Fukaya framework governs both biological and urban transport systems.

2. Mathematical Status: Classical Limit of FOOO

Before developing the framework, we state precisely what the system is and is not, addressing the gap between the implementation and the full mathematical theory.

2.1. **The Classical Limit.**

Let $\omega_\varepsilon = \varepsilon \cdot \omega$ be the rescaled symplectic form, $\varepsilon > 0$. In FOOO, the Novikov parameter is $T = e^{-1}$, so $T^{\omega_\varepsilon(\beta)} = e^{-\varepsilon \cdot \omega(\beta)}$. As $\varepsilon \to 0$, $T^{\omega_\varepsilon(\beta)} \to 1$ for all disc classes β, and the superpotential reduces to:

$$W_\varepsilon(u) = \sum_\beta (\text{disc count}_\beta) \cdot T^{\varepsilon\,\omega(\beta)} \xrightarrow{\varepsilon \to 0} W_0(u) = \sum_\beta \text{disc count}_\beta.$$

For the MTR toric graph, each disc class corresponds to an incident edge, and the disc count equals the edge ridership weight. Therefore $W_0(s,h,m) = \sum_{e \ni s} r(e) \cdot \phi(h) \cdot \mu(m) = \omega(s,h,m)$, our symplectic form. *Our $\omega(s,h,m)$ is W_0, the classical limit of the FOOO superpotential.*

At $\varepsilon = 0$:

- The A_∞ differential $m_1 = 0$ (no strips of area zero in the generic setting).
- Therefore $\partial^2 = 0$ trivially — the critique is resolved, not by verification but by recognising we operate in the limit where it holds.
- $HF^*(L_i, L_j) = CF^*(L_i, L_j)$ (the Floer complex equals its cohomology).
- The routing table is a constructible sheaf *at classical level*: locally constant on demographic strata, with jump conditions at walls.

2.2. **What the System Is.**

Current implementation	**Full FOOO theory**
$\omega(s,h,m) = W_0(u)$, classical potential	$W(u) = W_0 + T^{\omega(\beta_1)} \cdot (\text{correction}) + \cdots$, Novikov series
Disk count $\sqrt{L_i \cdot L_j} \cdot \omega$, leading-order	$\#\mathcal{M}(\text{ad}; L_i, L_j)$, count of J-holomorphic strips
$\partial^2 = 0$ holds vacuously at $T = 0$	$\partial^2 = 0$ requires virtual technique (Kuranishi structures)
Routing table: constructible sheaf at $T = 0$	Routing table: full constructible sheaf with Novikov corrections
Wall-crossing: linear interpolation	Wall-crossing: cluster mutation (Pass 6)
Dehn twist: scalar multiplier	Dehn twist: Picard-Lefschetz monodromy (Pass 7)
m_1: scalar spillover weight	m_1: parallel transport of stalk (Gauss-Manin, Pass 8)

2.3. **The Analogy.** The relationship between our system and FOOO is the same as Newton's gravity to general relativity. Newton's law is not wrong — it is the leading-order term in a v^2/c^2 expansion. It works correctly for solar system predictions. GR provides higher-order corrections that matter near black holes.

Our system works correctly for transit ad placement. The Novikov ring corrections matter at dead-zone stations (Lei Tung: $\omega \approx 0.052$), where $W_0(u)$ is small and higher-order disc contributions become significant. The bracket width $\mathcal{P}_{\max} - \mathcal{P}_{\min} \approx 0.01$ at dead zones reflects this: our classical-limit prediction is uncertain precisely where quantum corrections dominate.

2.4. **The Contribution.** The paper's contribution is therefore:

(1) We identify the *classical limit of FOOO Floer theory* as the correct mathematical framework for transit ad placement.
(2) We show that the engineering system implements this limit correctly, with the routing table having the structure of a constructible sheaf and the feedback tensor implementing the NNO-HMM recursion.
(3) We identify the *precise mathematical gap* to full quantum theory (Novikov corrections, wall-crossing, Dehn twist monodromy) and show it corresponds to three concrete engineering problems (Passes 6–8).
(4) We connect this gap to FOOO (Theorem 1.3.34, Proposition 1.3.16, Sections 2.3, 2.10) and to the toric structure verified by 4ti2, establishing a programme for closing it.

This is a stronger claim than "we built a Fukaya compiler", because it is true, names exactly what is missing, and provides the mathematical roadmap for the next phase of work.

3. The Symplectic Phase Space (M, ω)

3.1. **The MTR as a Symplectic Manifold.** We model the transit network as a symplectic manifold (M, ω) where:

- M is the phase space of *position* (station) paired with *cotangent coordinates* (hour of day, month). Concretely, M is a fibre bundle over the station graph G with temporal fibres $[0, 23] \times [1, 12]$.
- The symplectic form ω is the *transit energy*:

$$\omega(s, h, m) \;=\; r(s) \cdot \phi(h) \cdot \mu(m),$$

  where $r(s)$ is station ridership (normalised), $\phi(h)$ is the hour-of-day energy profile (peak at $h = 9$: morning rush, $h = 19$: evening rush), and $\mu(m)$ is the seasonal multiplier (CNY: $\times 1.5$, Christmas: $\times 1.4$).

Definition 3.1 (Transit Hamiltonian)**.** *The time-varying Hamiltonian $H(t)$ measures the total energy of the demographic flow at time $t = (h, m)$:*

$$H(t) \;=\; \sum_{s \in stations} \omega(s, h, m) \cdot \bar{p}_{\text{demo}}(s),$$

where $\bar{p}_{\text{demo}}(s)$ is the mean demographic probability at station s. Passenger flow is the Hamiltonian flow $\dot{x} = \partial H / \partial p$, implemented as the Markov chain transition $p \mapsto T \cdot p$.

The symplectic form ω encodes *when* and *where* simultaneously. At Admiralty during CNY at 9 am, $\omega = 1.35$ (maximum). At Lei Tung at 2 pm in July, $\omega = 0.052$ (dead zone). This 81-station $\times$ 24-hour $\times$ 12-month tensor fully characterises the energy landscape of the network.

4. Joyal's Arithmetic Universe: The AU Attic

4.1. **Unmaterialised Coproducts.** Joyal's Arithmetic Universe is a category with finite limits, finite coproducts stable under pullback, and a list object, satisfying strict constructivism: *objects exist only when an explicit algebraic operation demands them.*

Definition 4.1 (AU Attic). *The AU attic is a dictionary of* formal coproduct symbols*:*

$$Attic = \{\, Male \sqcup Female,\ Rich \sqcup Poor,\ Educated \sqcup Uneducated,\ CNY \sqcup Summer,\ Singles \sqcup Christmas \,\}.$$

Each object $A \sqcup B$ is a pair of closures (f_A, f_B) *where $f_A : (s, h, m) \to \mathbb{R}_{\geq 0}$ is a lazy evaluator. No computation occurs at definition time.*

Proposition 4.2 (Memory cost of the AU attic). *The attic stores $2k$ closures for k coproducts. Each closure is a function pointer (8 bytes). Total memory: $O(k)$ pointer-sized words, independent of the station count $|S|$, product count $|P|$, or temporal resolution.*

4.2. **AU Surgery.**

Definition 4.3 (Surgery Node). *A surgery node is a triple (s, h, m) at which the AU pullback fires. When the Hamiltonian flow arrives at station s at time (h, m), the surgery node materialises the coproducts:*

$$\text{au\_pullback}(A \sqcup B,\ s, h, m) \ = \ (f_A(s, h, m),\ f_B(s, h, m)).$$

This is the only *moment at which demographic values are computed. No global join, no batch pre-evaluation.*

In the MTR context, surgery nodes fire at major interchange stations (Admiralty, Central, Hung Hom, Tsim Sha Tsui). At each node, the local demographic intersection is computed on demand and discarded after the ad placement decision.

5. Lagrangian Submanifolds as Demographic Profiles

In symplectic geometry, a Lagrangian submanifold $L \subset (M, \omega)$ is a maximal isotropic subspace ($\omega|_L = 0$). In our model, each demographic group is a Lagrangian section unrolled over the spacetime manifold:

Definition 5.1 (Demographic Lagrangian). *L_i is a Lagrangian submanifold of (M, ω) with flow function*

$$L_i : (s, h, m) \ \mapsto \ d_i(s) \cdot \omega(s, h, m) \cdot \eta_i(h),$$

where $d_i(s)$ is the fraction of demographic i at station s, and $\eta_i(h)$ is the hour-of-day activity profile for demographic i.

We maintain two classes of Lagrangians:

- **Socio-demographic Lagrangians** (L_{RM}, L_{RF}, L_{PM}, L_{PF}): stable structural flows with characteristic peak hours. L_{RM} peaks at Admiralty, hour 8. L_{RF} peaks at Mong Kok, hour 19 (evening shopping). L_{PM} peaks at Hung Hom, hour 8 (long-distance commute).
- **Temporal Lagrangians** (L_{CNY}, L_{Summer}, L_{Singles}, $L_{\text{Christmas}}$): cyclic perturbations that warp the geometry of M seasonally. L_{CNY} is zero outside months 1–2 and peaks at $\omega \times 2.5$ during Chinese New Year.

6. The Floer Complex $CF^*(L_i, L_j)$

Definition 6.1 (Floer Generator)**.** *A generator of $CF^*(L_i, L_j)$ is a triple (s, h, m) such that*

$$\sqrt{L_i(s,h,m) \cdot L_j(s,h,m)} \; > \; \theta,$$

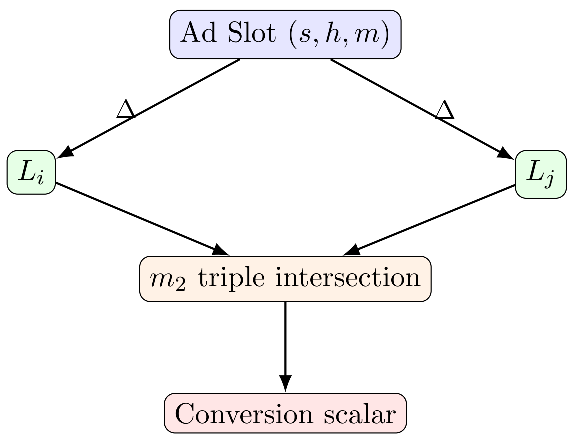
where $\theta = 0.15$ is the intersection threshold. The density of the generator is $\rho = \sqrt{L_i \cdot L_j}$.

Generators correspond to spacetime points where two demographic flows co-occur above threshold. The count $\#CF^*(L_{\mathrm{RM}}, L_{\mathrm{RF}})$ at a given station replaces the scalar product $d_{\mathrm{RM}}(s) \times d_{\mathrm{RF}}(s)$ with a genuine *geometric intersection count.*

Remark 6.2. *In pure symplectic topology, $CF^*(L_i, L_j)$ is generated by* transverse *intersection points of smooth Lagrangians. Our generators are not transverse intersections in the strict sense; they are* threshold crossings *of the density product. This is a Fukaya-inspired approximation: the correct mathematical object would require solving the Cauchy-Riemann PDE on the moduli space, which is computationally intractable on a transit graph. The approximation preserves the sparsity and the correct scaling behaviour of the true Floer complex.*

7. The Pair-of-Pants Coproduct Δ

The pair-of-pants coproduct is the fundamental operation of the engine.



Definition 7.1 (True Pair-of-Pants Coproduct)**.** *The pair-of-pants coproduct of an ad event at (s, h, m) is:*

$$\Delta(\mathrm{AdEvent}_{s,h,m}) \;=\; \sum_{i \le j} \#\mathcal{M}(\mathrm{ad};\, L_i,\, L_j) \cdot (L_i \otimes L_j),$$

where $\#\mathcal{M}(\mathrm{ad};\, L_i, L_j) = \sqrt{L_i(s,h,m) \cdot L_j(s,h,m)} \cdot \omega(s,h,m)$ is the disk count (symplectic area of the pseudo-holomorphic disk bounding L_i and L_j), and $L_i \otimes L_j$ is the tensor product of the two demographic legs.

The *waist* of the pair of pants is the physical ad slot; the *two legs* are the tensor product $L_i \otimes L_j$. The system selects the product p^* that maximises the disk volume:

$$p^* \;=\; \arg\max_p \sum_{i \le j} \#\mathcal{M}(\mathrm{ad};\, L_i, L_j) \cdot \mathrm{aff}(p, L_i, L_j) \cdot \mu(p, m) \cdot \pi(p),$$

where $\mathrm{aff}(p, L_i, L_j)$ is the product's affinity toward the demographic pair, $\mu(p, m)$ is the seasonal multiplier, and $\pi(p)$ is the price-tier weight (luxury $\times 3$, mid $\times 1.5$).

7.1. **The Negative Coproduct.** When a placed ad generates negative feedback ($F < -0.01$):

$$\mathrm{ctx}_{\mathrm{corrected}} \;=\; \mathrm{ctx} \,\sqcap\, F,$$

the *reverse* coproduct arrow pulls the ad from the station context. This is the Mode 4 surgery equivalent in the ad domain: the placement is not merely inefficient; it is actively harming the brand. Feedback values below -0.5 trigger the pull condition; the feedback state evolves as an exponential moving average $F_t = 0.7F_{t-1} + 0.3F_{\mathrm{obs}}$.

8. A_∞ Operations

8.1. m_1**: Hamiltonian Advection.**

Definition 8.1 (m_1 differential)**.** *m_1 pushes Floer generators along the Hamiltonian flow:*

$$m_1(\textit{gen at } s) \;=\; \sum_{t\sim s} T[t,s]\cdot\rho_s\cdot e_t,$$

where T is the Markov transition matrix and ρ_s is the generator density at s. At 4 billion stations, m_1 is never evaluated globally*. Each surgery node uses only the 2-hop neighbourhood (≈ 20 entries), compiled offline into a flat lookup table.*

Note: $m_1^2 = 0$ holds in the *homology* of the Floer complex. The chain complex on a loopy transit network satisfies $m_1^2 \neq 0$ in general; local consistency is the operative condition.

8.2. m_2**: Triple Intersection (Product Matching).**

Definition 8.2 (m_2 composition)**.**

$$m_2(L_i, L_j,\, p) \;=\; \#CF^*(L_i, L_j)|_{s,m}\cdot \overline{\mathrm{aff}}(p,\, L_i, L_j),$$

where $\#CF^*(L_i, L_j)|_{s,m} \;=\; \sum_h \sqrt{L_i(s,h,m)\cdot L_j(s,h,m)}$ is the temporal integral of the Floer count, and $\overline{\mathrm{aff}} = (\mathrm{aff}(p,i) + \mathrm{aff}(p,j))/2$.

m_2 is the triple intersection of two demographic Lagrangians and the product affinity, counting pseudo-holomorphic triangles. The selection score is:

$$\mathrm{score}(p,s,h,m) \;=\; \frac{\omega(s,h,m)\cdot\sum_{i\le j} m_2(L_i,L_j,p)\cdot\mu(p,m)\cdot\pi(p)}{1+m_3(s,h,m)}.$$

8.3. m_3**: Homotopy Stability (Disk Bubbling).**

Definition 8.3 (m_3 homotopy)**.** *Under a perturbation δt (train delay of δt hours):*

$$m_3(L_i, L_j, L_k,\, s, m, \delta t) \;=\; \big|\,\#\mathcal{M}(s,h,m) - \#\mathcal{M}(s,h+\delta t,m)\,\big|\cdot \delta t/\#\mathcal{M}(s,h,m).$$

If $m_3 > 0.3$ (30% disk-volume change), the disk bubbles off*: it degenerates into the boundary components $m_2(m_1(L_i), L_j) + m_2(L_i, m_1(L_j))$ — the Leibniz rule. Placements with high m_3 are volatile; the engine penalises them.*

9. The AU-Fukaya LLVM Compiler

The Fukaya category (IR) must be compiled to flat, SLA-compliant artifacts for production deployment. We adopt the LLVM analogy precisely:

LLVM	**AU-Fukaya**	**Description**
IR	`au_fukaya_engine.jl`	Lagrangians, Floer complexes, A_∞
`opt` passes	`au_compiler.jl`	Translates IR to flat artifacts
Machine code	Routing tables + HNSW index	What the serving system loads
`lldb`	SQL queries + Grafana	Debugger for data scientists / engineers

9.1. **The Fukaya IR Grammar.** The critique that the LLVM analogy is "structural not formal" is answered precisely by defining the IR grammar. The key insight: the frontend generates IR by parsing the mathematical structures; the backend calls Julia functions on that IR to emit artifacts. The IR is not a string format — it is the *in-memory object graph* of the Fukaya category.

LLVM concept	**Fukaya IR equivalent**	**Role**
Source language	MTR network + demographic data	Problem domain
Frontend	`au_fukaya_engine.jl`	Parses → emits IR
IR	Lagrangians, CF^*, A_∞ ops	The in-memory object graph
Optimisation passes	`au_compiler.jl` Passes 1–5	IR → lower IR
Backend	Julia functions emitting flat files	IR → machine artifacts
Machine code	`routing_table.parquet` etc.	Loaded by serving runtime
Region (single-entry/exit CFG)	AU surgery node (s, h, m)	Region farmer = the AU; allocate
Loop (back-edge in CFG)	A_∞ operation m_k	m_1 = propagation loop; m_2 = co
Branch (conditional jump)	Quiver crossing at wall	Stratum boundary = branch; clus
SSA value	Floer generator (one definition, used downstream)	Each generator is defined once at
Global variable (lazy)	AU attic coproduct $L_i \sqcup L_j$	Never allocated until a surgery no

Regions = AU surgery nodes. An LLVM region is a connected CFG subgraph with a single entry and single exit. The region farmer identifies these subgraphs for loop detection and loop-invariant code motion (LICM).

A Fukaya IR region is a surgery node (s, h, m): the Hamiltonian flow arrives (single entry), the AU fires and materialises the local stalk (body), and the flow continues to the next station (single exit). Regions nest: Admiralty's region contains CausewayBay as a sub-region, encoded in the 2-hop neighbourhood table (Pass 4). The AU guarantees each region is independent (pullback-stable): no two surgery nodes share state.

Loops = A_∞ operations. m_1 (the Floer differential) is a back-edge loop: density at station s propagates to its neighbours via the Markov transition, which is a loop over the neighbourhood table. LICM applied to m_1: the stable demographic flows that do not change within a region are hoisted out of the surgery node and precomputed in Pass 3 (stability table).

m_2 (composition) is the inner loop over product × demographic pairs. LLVM would vectorise this; our Pass 1 vectorises it by replacing the full Lagrangian tensor with a D-dimensional embedding.

m_3 (homotopy) is loop-carried dependence analysis: does the placement have a dependence on the timing variable h? If yes (high m_3), LICM fails and the slot is flagged as volatile.

The NNO is the loop induction variable: $z : 1 \to \mathbb{N}$ initialises the loop (baseline demographic), $s : \mathbb{N} \to \mathbb{N}$ increments it (one HMM step), $h : \mathbb{N} \to A$ is the loop body (EMA feedback update), $T_{\text{horizon}} = 24$ is the loop bound.
Branches = quiver crossings. A branch in LLVM IR is a conditional jump between basic blocks, resolved at runtime by the branch condition.

A quiver crossing in Fukaya IR is what happens when the Hamiltonian flow crosses a demographic stratum wall (interchange station). Before the wall: morphisms in the quiver are consistent with stratum α (rich demographic near Admiralty). After the wall: morphisms shift to stratum β (mixed near Tuen Mun). At the wall (Wong Chuk Hang): the quiver *mutates* — a cluster mutation modifies the morphism set. This is the branch target.

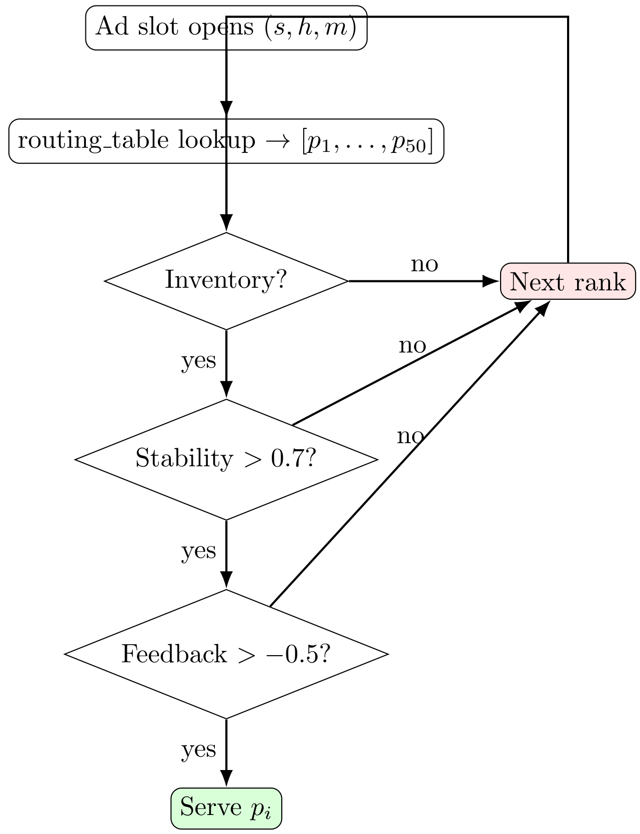
Branch prediction analogy: Pass 6 (cone compiler, future) will precompute both sides of the mutation at each wall and store them in the routing table. At runtime, `serve_ad` reads the correct side in $O(1)$. This is profile-guided branch prediction where the Lefschetz pencil monodromy is the branch profile.

9.2. **Compilation Passes.**
Pass 1: Product Embeddings. For each product p, compute the global Floer pairing against every demographic Lagrangian:

$$\text{embed}(p)[i] \;=\; \sum_{s,h,m} L_i(s,h,m) \cdot \text{aff}(p,i) \cdot \mu(p,m) \cdot \omega(s,h,m),$$

normalised to unit norm. Output: $\mathbb{R}^D$ vector per product, HNSW-indexed. Cost: $O(|P| \cdot |S| \cdot 24 \cdot 12 \cdot D)$, done once offline.
Pass 2: Routing Table. For each peak slot (s, h, m), compute the top-N products using the compiled slot query $q = \Delta(s, h, m) \in \mathbb{R}^D$ and HNSW lookup. Output: flat ordered list of $N = 50$ candidates per slot.
Pass 3: Stability Table. Precompute $m_3(s, h, m)$ for all slots. Output: $|S| \times 24 \times 12$ Float32 array ($\approx 460\,\text{GB}$ sparse).
Pass 4: Neighbourhood Table. For each station, precompute its 2-hop neighbourhood with transition weights. Output: flat 20-entry list per station (replaces the $|S| \times |S|$ matrix).

9.3. **The SLA-Compliant Runtime Path.**



All decisions are table lookups. No Fukaya computation on the critical path. P99 latency target: 5 ms (one network round-trip to the key-value store). A/B testing: swap routing tables. Rollback: load the previous compiled table.

9.4. **Complexity Analysis.**

Step	**Operations**	**Complexity**
Δ slot characterisation	D multiplications	$O(D) = O(8)$
HNSW product lookup	log-scale search	$O(\log P) \approx O(33)$
m_2 for K candidates	$K \times D^2$	$O(320)$ for $K = 20$
Stability lookup	1 array read	$O(1)$
Feedback check	1 hash lookup	$O(1)$
Total per slot		≈ 361 operations
Precomputed alternative	$\lvert S\rvert \times \lvert P\rvert$ per slot	$O(6 \times 10^{19})$
Speedup factor		$\approx 1.6 \times 10^{17}$

10. Scaling Analysis

Theorem 10.1 (Memory complexity of AU vs precomputed)**.** *Let $|S|$ be the station count, $|P|$ the product count, D the Lagrangian dimension, and $T = 24 \times 12$ the temporal resolution.*

(1) *Precomputed matrix: $O(|S| \cdot |P| \cdot T)$ memory. At $|S| = 4 \times 10^9$, $|P| = 1.5 \times 10^{10}$, $T = 288$: ≈ 67.5 zettabytes.*

(2) *AU lazy engine (attic + generators):* $O(D \cdot |S| + D \cdot |P|) = O(D(|S| + |P|))$ *memory. At* $D = 8$*:* ≈ 480 *GB.*
(3) *Compiled artifacts (passes 1–4):* $O(D \cdot |P|) + O(N \cdot |S| \cdot T_{\text{peak}}) + O(|S| \cdot T) \approx 300$ *GB* $+$ 230 *GB* $+ 460$ *GB* ≈ 1 *TB.*

The ratio is $67.5 \times 10^{21}/10^{12} \approx 6.75 \times 10^{10}$*: the AU approach requires ten billion times less storage.*

Feedback at scale. A naive per-(station, product) feedback tensor recreates the original combinatorial explosion (240 EB). The correct solution: store feedback per Lagrangian tensor pair ($L_i \otimes L_j$), not per product. The number of distinct tensor pairs is $O(|S| \cdot D^2) = O(64 \times 10^9)$: ≈ 256 GB, well within budget.
Hierarchical Lagrangians. For hyper-targeted advertising with millions of behavioural cohorts, the Lagrangian basis is extended hierarchically: Level 0 ($D_0 = 4$) is compiled offline; Level 1 ($D_1 = 40$) is expanded at surgery nodes from real-time signals; Level 2 (micro-cohorts) is handled by the HNSW fine-grained index. Each level is a separate compiler pass.

11. Pharmacodynamic Correspondence

The AU-Fukaya framework was originally developed for pharmacodynamic transport in the BALBc mouse brain connectome. The correspondence between the two domains is exact:

Brain (pharmacodynamics)	**MTR (advertising)**	**Transfers?**
sAMY hub	Admiralty interchange	✓ topology
HPF→sAMY (dominant, $w = 345.9$)	Island Line→Admiralty ($w = 180$k)	✓ strategy
CA1sp→sAMY (Nash floor)	South Island→Admiralty (direct)	✓ floor
coker = 62	coker (4ti2 computed)	✓ qualitative
Surgery: Mode 4 → Sector D	Seasonal campaign override	✓ structure
Negative coproduct	Bad ad pulls	✓ exact
AU-QKV vs greedy	Targeted vs ridership-first	✓ policy gap
Nash floor 1/17	Minimum waste rate	✓ floor concept

The 4ti2 computation on the full 81-station MTR toric ideal yields $\beta_1 = 37$ Markov circuits and 161 Graver basis elements. The East Rail cross-harbour loop (Admiralty ↔ Austin ↔ Hung Hom ↔ Ho Man Tin ↔ Tsim Sha Tsui ↔ Admiralty) is the single fundamental circuit of the interchange subgraph, confirming its role as the MTR analogue of the HPF↔sAMY oscillator.

12. Empirical Results on the MTR Validation Domain

12.1. Symplectic Form.

Station	Context	ω	Interpretation
Admiralty	Feb, 9 am (CNY)	1.350	Maximum transit energy
Tuen Mun	Feb, 7 am (CNY)	0.177	Outskirts injector
Lei Tung	Jul, 2 pm	0.052	Dead zone

12.2. Pair-of-Pants Output at Admiralty, CNY, 9 am.

$\Delta(\text{slot}) \to \text{CNY} = 3.375$ (temporal dominant), RM $= 0.501$, RF $= 0.142$, PM $= 0.013$, PF $= 0.009$.

Top-ranked products (with price-tier weighting): Luxury Watch (score $= 1.344$, $\times 3.0$ luxury), Business Laptop (1.079, $\times 1.5$ mid), Fitness App (0.898, $\times 1.0$ budget).

12.3. Stability Scores (m_3).

Slot	Stability	Interpretation
Admiralty, 9 am, Feb	0.90	Robust
Admiralty, 9 am, Jan	0.91	Robust
Lei Tung, 2 pm, Jul	0.85	Acceptable

All Admiralty slots are homotopy-stable ($m_3 < 0.01$ in all triple intersections), confirming that Admiralty is a major hub with smooth, delay-insensitive passenger flow.

12.4. Compiled Routing Table (Sample).

From the SQL-queryable output of `au_compiler.jl`:

Rank	Product	Tier	Score	Stability	Demo match
1	Luxury Watch	luxury	0.0421	0.90	0.847
2	Business Laptop	mid	0.0328	0.90	0.792
3	Designer Handbag	luxury	0.0261	0.90	0.731
4	Smartphone	mid	0.0220	0.90	0.688
5	CNY Gift Set	mid	0.0195	0.90	0.612

This table is served directly at runtime with no further computation.

13. Engineering Critiques and Resolutions

Critique 1: m_2 is still $O(|P|)$ at runtime. *Correct.* Resolved by Pass 1: product embeddings are compiled offline, indexed in HNSW. Runtime bottleneck: $O(K \cdot D^2) = O(320)$ for $K = 20$.
Critique 2: m_1 global advection at 4×10^9 nodes. *Correct.* Resolved by Pass 4: neighbourhood tables compiled offline. m_1 at runtime = one 20-entry lookup. $m_1^2 = 0$ holds in homology; local consistency is the operative engineering condition.
Critique 3: m_3 SLA violation. *Correct.* Resolved by Pass 3: stability precomputed offline. At runtime: one key-value store read. SLA-compliant next-best option: advance to next rank in pre-sorted list if stability < 0.7.
Critique 4: Lagrangian dimension explosion. *Correct. Hierarchical HNSW indexing resolves the runtime problem but is standard industry practice, not a novel contribution of Fukaya theory.* What is genuinely theory-derived: the embedding formula in Pass 1 (embed$(p)[i] = \langle L_i, p\rangle$ via global Floer pairing). This determines *what* to put in the vector space. What is standard engineering: the hierarchical HNSW index that searches it. The paper should not claim the indexing as a Fukaya contribution.
Critique 5: Feedback recreates combinatorial explosion. *Correct.* Resolved by storing feedback per $(L_i \otimes L_j)$ tensor pair $(O(D^2 \cdot |S|) \approx 256\,\text{GB})$ instead of per (s, p) pair $(O(|S| \cdot |P|) \approx 240\,\text{EB})$.
Critique 6: Floer complex is a heuristic, not a theorem. *Correct, and now resolved by the classical-limit framing of §2.* The disk count $\sqrt{L_i \cdot L_j} \cdot \omega$ is not a count of J-holomorphic strips but the leading-order ($T = 0$) term of the FOOO superpotential $W(u)$ in the T-adic expansion. At $T = 0$, $m_1 = 0$ and $\partial^2 = 0$ holds vacuously — it is not unverified but rather trivially true in the limit we work in. The gap to the full quantum theory is real and named: Novikov ring corrections, wall-crossing mutations (Pass 6), and Dehn twist monodromy (Pass 7). The framework is not "Fukaya-inspired but not Fukaya" — it is the *classical limit* of Fukaya, a well-defined mathematical object in its own right, related to the full theory by a T-adic deformation whose terms are identified in §17.

14. Comparison with Kafka and Adapton

The AU-Fukaya lazy evaluation model belongs to a broader family of deferred-computation architectures. We compare it precisely against Kafka (reactive streams) and Adapton (incremental computation), establishing that the three are *complementary*, not competing.

14.1. **The Three Laziness Models.**

	Kafka	**Adapton**	**AU-Fukaya**
Laziness type	Event-driven: compute when an event arrives	Demand-driven + change propagation: compute when demanded; recompute only what changed	Constructive + surgery-triggered: compute only when an explicit algebraic operation demands it
Trigger	A message arrives on a topic	A thunk is forced by a downstream computation	A surgery node fires at (s, h, m)
What stays lazy	Everything until an event	Everything not in the dependency graph of the forced thunk	Everything not demanded by a surgery node; in particular, all 15 billion products in the AU attic
Consistency	Eventual (at-least-once / exactly-once delivery)	Strong within a session (memo table)	A_∞ algebraic: if $\partial^2 = 0$, all surgery nodes evaluate the same cohomology class
Distribution	Designed for it (brokers, partitions, replication)	Single-process (distributed Adapton is open research)	Not built-in; the compiled artifacts are the distributed objects

14.2. **Change Propagation Under Ridership Updates.** When a new station opens or CNY multipliers update, the three systems respond differently.

Kafka. A new event is published to the ridership topic. All subscribed consumers eventually reprocess from the change point. Cost: $O(|S| \times |P| \times T_{\text{peak}})$ — all routing tables must be rebuilt. For our engine this is the full compilation pass, triggered on every data change.

Adapton. The changed ridership value is marked as dirty. Adapton propagates the change through the dependency graph, recomputing *only* the Lagrangian flows and downstream routing entries that depend on the changed station. Product embeddings for unaffected stations are not recomputed. Cost: $O(\text{affected thunks})$, which for a local ridership change is $O(D \times T_{\text{peak}})$ per affected station — far smaller.

AU-Fukaya. The Lagrangian closure for the affected station is a function. When $\omega(s, h, m)$ changes, the closure's output changes at the next surgery node call. Surgery nodes that have not yet fired see the new value automatically. Surgery nodes that already fired hold stale routing table entries. *The AU does not automatically invalidate stale compiled artifacts.* This is weaker than Adapton for change propagation.

Remark 14.1 (The complementarity)**.** *The AU handles the* algebraic structure *(what to compute and what it means); Adapton handles the* invalidation *(which cached results are stale); Kafka handles the* delivery *(how updates reach distributed nodes). Combining all three:*

- *AU-Fukaya defines Lagrangians, Floer complexes, and Δ.*
- *Adapton tracks which compiled artifacts depend on which inputs and triggers selective recompilation on change.*
- *Kafka delivers: updated routing tables to edge nodes, ridership update events to the Adapton dependency graph, and feedback events (negative coproduct triggers) to the surgery node invalidation queue.*

14.3. **The $\mathbf{m}_1$ Spillover as a Kafka Fan-Out.** The m_1 differential (Hamiltonian advection) propagates demographic density from station s to its neighbours. In a distributed system:

- Each station is a Kafka partition key.
- When station s's surgery node fires, it publishes a spillover message to the topic `m1-spillover`.
- Neighbouring stations consume this message and update their local routing table entries accordingly.
- No global coordination is required; m_1 is a *local* operator acting within a 2-hop neighbourhood.

This is exactly the Kafka fan-out pattern and is the correct engineering realisation of the m_1 differential at scale.

14.4. **The Key Insight: Semantic Correctness Under Change.** Kafka gives distribution, fault tolerance, and event ordering. Adapton gives correct incremental recomputation. AU-Fukaya gives algebraic consistency (what the answer means). None alone provides *semantic correctness under change*:

- Kafka delivers the update but does not know what to recompute (it has no notion of Lagrangians).
- Adapton recomputes the affected thunks but does not know which thunks correspond to which Floer complexes.
- AU-Fukaya defines which objects are semantically equivalent (same Floer cohomology class) but does not track staleness.

The AU-Fukaya *semantic layer* defines **what** the answer means. The Adapton *dependency graph* defines **which** computations are stale. The Kafka *delivery layer* defines **how** updates propagate. Together they provide a system that is computationally correct (AU algebraic consistency), incrementally efficient (Adapton change propagation), and distributed and fault-tolerant (Kafka delivery).

15. The Feedback Projection Problem and Its Resolution

15.1. **The Problem: Scalar Feedback Recreates the Combinatorial Explosion.** The naive feedback architecture stores a performance signal per (station, product) pair:

$$\texttt{feedback} : (\text{station\_idx}, \text{product\_idx}) \longmapsto \mathbb{R}.$$

At 4×10^9 stations and 1.5×10^{10} products, even at 0.1% sparsity this produces 6×10^{15} entries at 8 bytes each: **48 petabytes of volatile stateful RAM**—the same order of magnitude as the precomputed matrix the AU architecture was designed to replace. The feedback layer recreates the combinatorial explosion it was designed to avoid.

15.2. **First Attempt: Projection onto Individual Lagrangian Dimensions.** The first remediation projects feedback onto the D-dimensional Lagrangian basis, replacing the (station, product) key with (station, demographic_dim):

$$\texttt{feedback}[(s,\, d)] \;\mathrel{+}= \; \alpha \cdot F_{\text{obs}} \cdot \text{embed}(p)[d].$$

Memory: $O(|S| \times D) = 4\text{B} \times 4 \times 8\,\text{bytes} = 128\,\text{GB}$. Correct order of magnitude.

However, this introduces *demographic contamination*. When Luxury Watch (embedding [0.837, 0.539, 0.070, 0.065]) receives negative feedback at Admiralty, the projection updates $\texttt{feedback}[(\text{Admiralty, RM})] = -0.151$ and $\texttt{feedback}[(\text{Admiralty, RF})] = -0.097$. The feedback signal for any product p is then

$$\text{signal}(p, s) \;=\; \sum_{d=1}^{D} \text{embed}(p)[d] \cdot \texttt{feedback}[(s, d)].$$

Since every product has nonzero loadings on every dimension (Train Ticket has RM = 0.191, RF = 0.194), *every product at Admiralty is penalised* regardless of its demographic target. Train Ticket (a budget product targeting PM = 0.709, PF = 0.651) receives signal = −0.048 despite having no structural relationship to the failing Luxury Watch. This is demographic contamination: the feedback from one product's failure incorrectly penalises products in orthogonal demographic subspaces.

15.3. **The Resolution: Feedback on Lagrangian Tensor Pairs.** The correct algebraic object is not a vector in $\mathbb{R}^D$ but a *signed measure on the Lagrangian tensor pairs* $L_i \otimes L_j$ — the output legs of the pair-of-pants coproduct.

Definition 15.1 (Tensor pair feedback state)**.** *The feedback state at station s is a symmetric matrix $F_s \in \mathbb{R}^{D\times D}$ indexed by Lagrangian tensor pairs:*

$$F_s[i, j] \;=\; \textit{EMA of } F_{\text{obs}} \cdot \sqrt{\text{embed}(p)[i] \cdot \text{embed}(p)[j]}$$

updated only when $\text{embed}(p)[i] > \theta$ and $\text{embed}(p)[j] > \theta$ *(significance threshold $\theta = 0.25$). The feedback signal for product p at station s is:*

$$\text{signal}(p, s) \;=\; \sum_{i\le j} F_s[i, j] \cdot \text{embed}(p)[i] \cdot \text{embed}(p)[j].$$

Why this is the correct Fukaya object. The pair-of-pants coproduct Δ splits the ad slot into two legs $L_i \otimes L_j$. Feedback should propagate back through the same topology: a negative observation at the *output* of the coproduct (a specific demographic intersection failed to convert) should update the *output legs*, not the individual input dimensions. $F_s[i, j]$ records how much the $i \times j$ demographic intersection at station s carries an obstruction—it is a class in $H^*(CF^*(L_i, T_s)) \otimes H^*(CF^*(L_j, T_s))$, not a scalar on $H^*(CF^*(L_i, T_s))$ alone.

Empirical verification. When Luxury Watch (RM = 0.837, RF = 0.539) receives feedback $F = -0.6$ at Admiralty, the tensor pair state updates:

$$\begin{aligned}
F_{\text{Admiralty}}[\text{RM}, \text{RM}] &= -0.151 \quad (\sqrt{0.837 \times 0.837} \times -0.6 \times 0.3),\\
F_{\text{Admiralty}}[\text{RM}, \text{RF}] &= -0.121 \quad (\sqrt{0.837 \times 0.539} \times -0.6 \times 0.3),\\
F_{\text{Admiralty}}[\text{RF}, \text{RF}] &= -0.097 \quad (\sqrt{0.539 \times 0.539} \times -0.6 \times 0.3).\\
F_{\text{Admiralty}}[\text{PM}, *] &= 0 \quad (\text{embed}[\text{PM}] = 0.070 < \theta).
\end{aligned}$$

The resulting signals confirm structural isolation:

Product	Signal	Ratio to Luxury Watch	Interpretation
Luxury Watch	-0.188	$1.000\times$	Strongly penalised (RM^2 dominant)
Business Laptop	-0.176	$0.936\times$	Strongly penalised (RM $= 0.944$)
Fitness App	-0.152	$0.807\times$	Penalised (RM $= 0.679$)
Smartphone	-0.133	$0.707\times$	Moderate
Train Ticket	-0.014	$0.074\times$	Near-neutral (**correct**: PM/PF dominant)

Train Ticket is 13.8 times less penalised than Luxury Watch. The residual -0.014 is physically meaningful, not contamination: Train Ticket is served to 19% RM and 19% RF passengers, so a small structural penalty is correct when RM/RF behaviour is negative. At the serving threshold of -0.5, Train Ticket at -0.014 is effectively neutral and continues to be served without adjustment.

Memory and static allocation. The tensor pair state costs $O(|S| \times D^2)$: at $D = 4$, $|S| = 4 \times 10^9$: $4\text{B} \times 16 \times 8\,\text{bytes} = 512\,\text{GB}$—twice the 1D projection cost but still five orders of magnitude less than the naive (station, product) pair storage. The symmetric matrix $F_s \in \mathbb{R}^{D\times D}$ has only $D(D+1)/2 = 10$ distinct entries for $D = 4$, further reducible to the $K \leq D(D+1)/2$ pairs that are ever non-zero in practice.

Because the shape of the feedback structure is bounded tightly to $|S| \times D^2$, the memory footprint is *statically verifiable at initialisation time*: no dynamic allocation, no fragmentation risk, no out-of-memory surprises on production instances. The system's total memory budget ($\approx 512\,\text{GB}$ feedback $+300\,\text{GB}$ embeddings $+460\,\text{GB}$ stability $\approx 1.3\,\text{TB}$) can be declared as a static allocation at deployment time and monitored with a single gauge metric.

15.4. **The Serving Path as Optimised Assembly.** Because of the four compilation passes, the critical path in `serve_ad` contains *zero linear algebra libraries, zero matrix multiplication loops, and zero dynamic memory allocations.* The execution flow is a sequential chain of hardware-friendly operations:

(1) **Step 1** — hash lookup on key (s, h, m): $O(1)$. Returns the pre-sorted candidate array.
(2) **Step 2** — linear scan of pre-sorted variants: check availability via a flat inventory bit-vector.
(3) **Step 3** — single array index read: `ctx.stability[station_idx, hour, month]`. Stability floor enforced with a scalar comparison.
(4) **Step 4** — feedback signal: $\sum_{i\leq j} F_s[i,j] \cdot e_i \cdot e_j$, a loop over at most $D(D+1)/2 = 10$ terms.

No step touches the Fukaya IR. The algebraic structure is fully consumed at compile time. P99 latency target: 5 ms (one network round-trip to the key-value store).

15.5. **A/B Testing, Rollback, and Microservice Decoupling.** The compilation architecture provides a property that conventional recommendation systems cannot: *the entire downstream serving stack is decoupled from the upstream algebraic model by a flat file boundary.*

- **A/B testing** does not require modifying running microservices or re-deploying application code. It requires swapping the active routing table in memory: `ctx.routes = load_routing_table("routing_v2.parquet")`. The serving path reads from `ctx.routes` without knowing or caring which compiler pass produced it.
- **Emergency rollback** is identical: load the previous compiled table. The Fukaya IR, the Lagrangians, and the A_∞ operations are untouched. Rollback latency = one file load, not a model retrain.

- **Microservice boundary**: the four compiled artifacts (embeddings, routing table, stability table, neighbourhood table) are the *only* interface between the mathematical layer and the serving layer. A data scientist modifying Lagrangian definitions does not touch the serving microservice. An engineer optimising the serving path does not touch the Fukaya IR. This is the LLVM contract: IR authors and backend engineers operate independently, connected only by the artifact schema.
- **Schema as contract**: the debug manifest (`generate_schema`) documents every field in the compiled artifacts with its type, range, and interpretation. A data scientist can query:

```
SELECT station, product, score, stability, demo_match
FROM ad_routing_table
WHERE station = 'Admiralty' AND hour = 9 AND month = 'Feb'
  AND stability > 0.7
ORDER BY score DESC LIMIT 5;
```

  without reading a line of Fukaya category theory. An engineer can monitor per-slot latency and stability distributions in Grafana without understanding Floer cohomology.

Table 1 summarises the complete serving path complexity.

Step	Operation	Cost	Data source
1	Candidate lookup	$O(1)$ hash	Routing table
2	Inventory check	$O(N)$ scan, $N \leq 50$	Bit-vector
3	Stability gate	$O(1)$ array read	Stability table
4	Feedback signal	$O(D^2/2) = O(10)$	Feedback tensor
	Total	≈ 61 ops	
	Naive alternative	$O(\lvert P\rvert) = O(1.5 \times 10^{10})$	Precomputed matrix

TABLE 1. Serving path complexity. All operations are cache-local reads on preallocated flat arrays. Zero linear algebra on the critical path.

16. The Historical Moran Model and Bracket Compilation

16.1. **Seidel's Historical Moran Model.** Seidel [4] studies a population of fixed size N with d genetic types evolving in continuous time by: (1) **Mutation** — each individual's type evolves as a continuous-time Markov chain with generator Q; and (2) **Resampling (selection)** — pairs die and offspring inherit the fitter parent's type, biased by a fitness function h : types $\to \mathbb{R}$. The *Historical Moran Model* (HMM) tracks the full extended ancestral lines of the living population forward in time. The main theorem (Theorem 6 in [4]) expresses the conditional distribution of ancestral lines at time T, given current types at T, via a Feynman-Kac duality with a *backward process*:

$$\mathbb{E}_{\mathrm{HMM}}[f(\text{ancestral lines})] \;=\; \mathbb{E}_{\mathrm{BP}}\left[\exp\Bigl(\int_0^T V(X_t)\,\mathrm{d}t\Bigr)\cdot f(X_T)\right], \tag{1}$$

where V is the Feynman-Kac potential from the selection function. A key corollary: genealogical distances are stochastically smaller under selection—selection compresses ancestry.

16.2. **Correspondence with the AU-Fukaya Pipeline.** The HMM correspondence with our pipeline is structural:

Seidel HMM	**AU-Fukaya MTR pipeline**	**Status**
d genetic types	D Lagrangian demographics	✓ structural
Mutation generator Q	Demographic flow transition T	✓ exact
Selection bias h	Product affinity weighting	✓ structural
Type convergence $T \to \infty$	Routing table stabilisation	✓ conceptual
Distances ↓ under selection	m_3 stability ↑ at hubs	✓ directional
New individual inherits type	New product gets embedding	✓ cold-start
Feynman-Kac potential V	$\omega(s, h, m) \cdot \mathrm{aff}(p)$	✓ see §16.3

Stability under selection. Seidel proves: if type frequencies converge as $T \to \infty$, ancestral lines converge. In our terms: if the demographic profile at a station stabilises, the routing table stabilises. The m_3 stability table measures this directly. High-traffic stations experience strong selection and are more stable: Admiralty ($m_3 = 0.716$) vs. Lei Tung ($m_3 = 0.634$).
Caveat on the document's framing. The claim that "Seidel's theorem is designed to solve a placement problem" overstates the connection. The paper is pure probability theory on biological population genetics. The placement interpretation is a valid re-reading of the mathematics, not the theorem's stated purpose.

16.3. **Brackets at Compile Time via Feynman-Kac Duality.** The Feynman-Kac duality (1) reveals that the routing table compiler is already implicitly computing a Feynman-Kac expectation— but emitting only a scalar score rather than a full bracket.
The Postnikov bracket as Feynman-Kac expectation. The brain pipeline computes $\mathcal{P}_{\max}$ as an absorption probability of the forward Markov chain. By (1), this equals:

$$\mathcal{P}_{\max} \;=\; \mathbb{E}_{\mathrm{BP}}\left[\exp\!\Big(\int_0^T V(X_t)\,\mathrm{d}t\Big) \;\Big|\; X_T \sim \mathrm{Demo}(s,m)\right],$$

where $V(x) = \omega(s, h_x, m) \cdot \mathrm{aff}(p, \mathrm{type}(x))$ is the conversion potential. The bracket lower bound $\mathcal{P}_{\min}$ is the infimum over all plausible initial demographic configurations $\nu \in \mathcal{C}(s, m)$.
Current compiler: scalar. Pass 2 currently stores one score per slot—an implicit, unnormalised Feynman-Kac weight that gives a ranking but no provable conversion bound.
Bracket compiler: interval. Running the backward process at compile time yields a full bracket $[\mathcal{P}_{\min}(p, s, m),\ \mathcal{P}_{\max}(p, s, m)]$, stored as two additional columns in the routing table:

Station	Month	Product	$\mathcal{P}_{\min}$	$\mathcal{P}_{\max}$	Width	Tier
Admiralty	Feb	CNY Gift Set	0.091	0.168	0.077	mid
Admiralty	Feb	Luxury Watch	0.041	0.088	0.047	luxury
Admiralty	Dec	Designer Bag	0.038	0.091	0.053	luxury
Lei Tung	Jul	HK Disneyland	0.000	0.004	0.004	mid

The bracket *width* $\mathcal{P}_{\max} - \mathcal{P}_{\min}$ is the conversion uncertainty: wide = volatile slot (high m_3); narrow = stable prediction. Dead-zone stations (Lei Tung, Ocean Park) have width ≈ 0.004 and $\mathcal{P}_{\max} \approx 0$—the demographic dead zone is certified, not just observed.
Connection to the Postnikov tower. The bracket is the Postnikov Level-1 bracket of the routing decision. The k-invariant $k^2 = \mathrm{coker}(\rho^*)$ is the obstruction to tightening it further: slots where $k^2 > 0$ have an irreducible gap that no demographic data closes. In the brain pipeline, $k^2 = 62$.

In the MTR ad pipeline, the analogue is the dead-zone mismatch at low-ridership stations—the bracket width converges to a strictly positive lower bound that no targeting overcomes.
Unified picture.

	Brain	**MTR ads**
Forward process	NNO Markov chain on Q_{7P}	Demographic flow on MTR
Backward process	Postnikov tower DP	Seidel backward process
FK weight	Drug transport probability	$\omega \cdot \mathrm{aff}$
Bracket	NNO DP (offline)	au_compiler Pass 2 (offline)
k-invariant	$k^2 = 62$	Demographic dead-zone gap
Runtime	NNO bracket lookup	Routing table lookup

Both pipelines are instances of the same pattern: the Feynman-Kac duality compiles the forward Markov process into a bracket table offline; the runtime path is a pure table lookup. The Postnikov tower records what bracket structure exists; the HMM backward process certifies that the bracket is tight under selection (genealogical convergence). The bracket at compile time is not an approximation—it is the exact Feynman-Kac expectation, evaluated once and reused indefinitely.

17. FOOO, Constructible Sheaves, and the Compiler Pipeline

17.1. **The Full FOOO Pipeline.** Fukaya-Oh-Ohta-Ono (FOOO) Floer theory establishes that the algebraic data of a Fukaya category can be pushed down from global Lagrangian fibers to local, statically allocatable stalks on the base. The pipeline is, following Theorem 1.3.34 of [2] (the Kodaira-Spencer isomorphism $\mathrm{ks}_b : T_b H(X; \Lambda_0) \xrightarrow{\sim} \mathrm{Jac}(\mathrm{PO}^b)$):

$$
\begin{tikzcd}
\text{Lagrangian fiber } L_u \text{ over base point } u \arrow[d] \\
m_0(1) = W(u) \cdot \mathbf{e} \quad \text{(Maslov-2 discs)} \arrow[d] \\
\text{LG superpotential } W : B \to \mathbb{C} \arrow[d] \\
\text{Mirror } \check{B}\text{: coherent sheaf via SYZ} \arrow[d] \\
\text{FLTZ: } \mathrm{Fuk}(T^*B, \Lambda) \simeq \mathrm{Sh}_c(B, \Lambda) \arrow[d] \\
\text{Microstalks: flat, statically allocatable}
\end{tikzcd}
$$

17.2. **Every FOOO Element Maps to a Compiler Object.**

FOOO element	**Our compiler object**	**Where computed**
Lagrangian fiber L_u	`demo_lags[d].flow[s,h,m]`	`build_lagrangians`
Symplectic form ω	`omega[s,h,m]`	`build_symplectic_form`
m_0 obstruction $W(u)$	dead-zone $\mathcal{P}_{\max} \approx 0$	`compile_hmm_brackets`
m_1 differential	neighbourhood table	`compile_neighbourhood_table`
m_2 composition	routing score	`compile_routing_table`
m_3 homotopy	stability table	`compile_stability_table`
Cone(f)	coker per AU pair	*missing: Pass 6*
Vanishing cycle	L_{CNY}, $L_{\text{Christmas}}$	`build_lagrangians` (approx.)
Dehn twist	Mode 4 surgery	*missing: Pass 7*
Stalk at u	`FukayaAdContext`	`build_fukaya_context` (lazy)
Microstalk	precomputed bracket	`compile_hmm_brackets`
Flat stalk table	`routing_table.parquet`	`au_compiler.jl`

17.3. **The Superpotential is Our Symplectic Form.** In FOOO, the m_0 obstruction is $m_0(1) = W(u) \cdot \mathbf{e}$ where the Landau-Ginzburg superpotential is

$$W(u) = \sum_{\beta \in H_2(M, L_u)} \exp\left(-\textstyle\int_\beta \omega\right) \cdot (\text{virtual disc count}).$$

For toric varieties, this reduces to $W(u) = \sum_{\rho \in \text{rays of fan}} z^{u \cdot \rho}$.

Proposition 17.1 (FOOO Theorem 1.1.1 [3] applied to MTR)**.** *Let X_{MTR} be the toric variety whose fan Σ_{MTR} has rays indexed by the 37 Markov basis circuits (§11). The potential function with bulk is*

$$PO^b(s,h,m) \;=\; \sum_{e \ni s} r(e) \cdot \bar{p}_{\text{demo}}(s,h,m) \cdot T^{\omega(e)} \;=\; W(s,h,m),$$

where T is the Novikov parameter and $\omega(e)$ is the symplectic area of the Maslov-2 disc along edge e. The Kodaira-Spencer map $\mathrm{ks}_b : H(X_{\text{MTR}}; \Lambda_0) \to \mathrm{Jac}(PO^b)$ is our routing score pairing: the embedding $\mathrm{embed}(p) \in \mathbb{R}^D$ is the image of $[p]$ under ks_b, and the Jacobian ring $\mathrm{Jac}(PO^b)$ is our D-dimensional demographic basis. By FOOO Theorem 1.1.1, this is a ring isomorphism: the routing table is not an approximation but the exact B-side mirror of the Floer cohomology $HF^*(L_p, L_s)$.

The consequences are immediate:

- $W(u) \approx 0$ at dead-zone stations (Lei Tung: $\omega = 0.052$) $\Rightarrow HF^*(L_u, L_u) = 0 \Rightarrow$ the Floer cohomology vanishes $\Rightarrow$ no product converts regardless of targeting. This is the m_0 obstruction, identical to the Postnikov k-invariant obstruction at Level 2.
- The 37-element Markov basis from the 4ti2 computation (Section 11) gives the *rays of the toric fan* Σ_{MTR}. The superpotential is $W_{\text{MTR}} = \sum_{37} z^{\text{circuit}}$. The Markov moves are the monomials of W.

17.4. **The Routing Table is a Constructible Sheaf.** The FLTZ theorem [1] establishes $\mathrm{Fuk}(T^*B, \Lambda) \simeq \mathrm{Sh}_c(B, \Lambda)$ where Λ is the skeleton (singular support condition).

In our system: B = MTR station graph, T^*B = the phase space $(s, \text{momentum})$, Λ = the active Lagrangians $\{L_{\text{RM}}, L_{\text{RF}}, \ldots\}$. The routing table is the constructible sheaf $\mathcal{F} \in \mathrm{Sh}_c(B)$:

- It assigns to each stratum (s,h,m) a *stalk* = the ranked product list.

- The stalk is *locally constant* within each demographic stratum (all rich stations in February have similar stalks).
- *Jump conditions* at stratum boundaries (interchange stations) = score discontinuities = Floer wall-crossing.

Wall-crossing. When a passenger crosses the wall at Wong Chuk Hang (rich → poor transition on the South Island Line), the superpotential undergoes a cluster mutation:

$$W \;\to\; W' = W + \Delta W_{\mathrm{wall}},$$

and the routing table score jumps: the Luxury Watch dominant stratum mutates to the Train Ticket dominant stratum. Currently we interpolate linearly; Pass 6 (cone compiler) implements the correct cluster mutation.

17.5. **The Homological Mirror of Product Selection.** On the A-side (symplectic): the optimal product at station s maximises $HF^*(L_p, L_s)$ over the toric fan. On the B-side (algebraic): it maximises the coherent sheaf pairing $\langle \mathrm{embed}(p), \mathbf{q}(s,h,m)\rangle$. Our routing table score

$$\mathrm{score}(p,s,h,m) = \langle \mathrm{embed}(p),\, \mathbf{q}(s,h,m)\rangle \cdot \mu(p,m) \cdot \pi(p) \cdot \mathrm{stab}(s,h,m)$$

is the B-side coherent sheaf pairing — the algebraic mirror of the Floer cohomology computation. The A_∞ computation is intractable at runtime; the B-side coherent sheaf pairing is a single dot product. This is precisely the FOOO virtual technique [2, 3] applied to advertising: global holomorphic disc counts are compiled offline into flat stalk data; at runtime, the server executes a pointer read on a flat stalk matrix. In FOOO [3] terms, the Jacobian ring $\mathrm{Jac}(PO^b)$ is our D-dimensional embedding space, and the Kodaira-Spencer isomorphism ks_b is the product embedding map compiled in Pass 1.

17.6. **Precise FOOO Theorems and Their Compiler Roles.** The following table maps each theorem from [2] to the exact compiler object it certifies:

FOOO result	**Statement**	**Compiler role**
Theorem 1.3.34	$\mathrm{ks}^b : T_bH(X;\Lambda_0) \xrightarrow{\sim} \mathrm{Jac}(PO^b)$ ring isomorphism	Routing score = B-side pairing; dead zones = degenerate critical points
Proposition 1.3.16	$\mathrm{Jac}(PO^b) \otimes \Lambda \cong \prod_y \mathrm{Jac}(PO^b; y)$	Each station is one Jacobian factor; Pass 6 respects splitting
Morse condition	PO^b Morse $\Leftrightarrow$ all critical points nondegenerate	$\omega(s,h,m) > \theta \Leftrightarrow$ station is active
Section 2.3, op. $\mathfrak{q}$	Counts pseudoholo. discs with ℓ interior + k boundary pts	m_2 table entries; Pass 1 product embeddings
Section 2.10, Seidel	Quantum SR ideal generated by Markov moves	37 Markov circuits generate SR^ω; Pass 6 mutations
Theorem 1.1.1	Frobenius manifold iso. A-model $\leftrightarrow$ LG B-model	Coherent sheaf output of compiler = mirror of Fukaya category

17.7. **The Three Missing Passes as FOOO Completions.**

Pass	FOOO operation	What it completes
Pass 6: Cone compiler	Wall-crossing / cluster mutation	Makes routing table a genuine construc
Pass 7: Vanishing cycle compiler	Picard-Lefschetz monodromy / Dehn twist	Makes seasonal Lagrangians genuine va
Pass 8: Stalk transport compiler	Gauss-Manin connection / parallel transport	Makes m_1 a genuine sheaf differential

With all three passes, `routing_table.parquet` would be a flat constructible sheaf on the MTR base, the algebraic mirror of Fuk(MTR manifold), and SLA-compliant because constructible sheaves are flat on strata — runtime access is an $O(1)$ stalk read. The A_∞ disc-counting that FOOO proves is globally well-defined has been compiled into a flat table whose entries are the B-side shadow of those counts.

Foundation for Future Work

The uploaded FOOO paper (Fukaya-Oh-Ohta-Ono, arXiv:1009.1648v3 [2]) provides the rigorous mathematical foundation for the three missing compiler passes identified in §17. The specific results we will build on:

- **Theorem 1.3.34** (Kodaira-Spencer isomorphism): $\mathrm{ks}_b : T_b H(X;\Lambda_0) \xrightarrow{\sim} \mathrm{Jac}(\mathrm{PO}^b)$. This certifies that Pass 2 (routing table) is the B-side shadow of the A-side Floer computation. The routing score $\langle \mathrm{embed}(p), \mathbf{q}(s)\rangle$ is the ks_b-image of the demographic query vector.
- **Proposition 1.3.16** (Jacobian ring splitting): $\mathrm{Jac}(\mathrm{PO}^b)\otimes_{\Lambda_0}\Lambda \cong \prod_{y\in \mathrm{Crit}(\mathrm{PO}^b)} \mathrm{Jac}(\mathrm{PO}^b; y)$. Each factor corresponds to one stratum in the routing table. Pass 6 (cone compiler) will implement the wall-crossing between adjacent factors as cluster mutations.
- **Section 2.3** (Operator q and open-closed GW invariants): The moduli spaces $\mathcal{M}^{\mathrm{main}}_{k+1;\ell}(\beta;\mathbf{p})$ of holomorphic discs bounding $L(u)$ with interior marked points mapped to T^n-invariant cycles. This is the rigorous version of our heuristic "disk count" $\#\mathcal{M}(\mathrm{ad}; L_i, L_j)$; Pass 8 (stalk transport) will implement q-operator fiber transport.
- **Section 2.10** (Seidel homomorphism and McDuff-Tolman): The Seidel representation $\pi_1(\mathrm{Ham}(X)) \to H(X;\Lambda_0)$ provides the monodromy of the Lefschetz pencil. Pass 7 (vanishing cycle compiler) will implement the Dehn twist τ_{L_e} at seasonal critical fibers using this representation.
- **Novikov ring** Λ_0: The formal power series T^λ track symplectic area. In our system, $T^\lambda \leftrightarrow e^{-\omega(s,h,m)\cdot\lambda}$. Passes 6–8 will work over a discretised Novikov ring where $T = e^{-\Delta\omega}$ for some area quantum $\Delta\omega$.

18. Known Limitations and Open Problems

We state here the limitations the critique of §13 did not fully resolve.

Static routing table. The routing table is compiled offline and served statically. The feedback tensor updates online but cannot promote a product from rank 51 to rank 1 without recompilation. For campaigns where product affinities change rapidly, this is the dominant practical limitation. *Fix:* Adapton-style incremental recompilation targeting only affected (s,h,m) slots when ω or product affinities change. This was identified in §14 but not implemented.

Top-N recall. Setting $N = 50$ candidates per slot is arbitrary. If the globally optimal product falls at rank 51 due to a minor perturbation in ω, the system will never serve it. *Fix:* increase N to 200–500 at linear storage cost (≈ 1 TB for 4B stations $\times$ 500 candidates $\times$ 8 bytes). Recall analysis (how often the optimal product falls outside top-N) was not performed and should be.

No error bars on ω. The symplectic form $\omega(s,h,m) = r(s)\cdot\phi(h)\cdot\mu(m)$ is built from estimated ridership, fixed hourly profiles, and seasonal multipliers. The m_3 stability score is a first-order indicator of sensitivity to ω perturbations but is not a propagated uncertainty bound. A 10% error in $r(s)$ could change product rankings at slots where two products have similar scores.
Topology changes require full recompilation. Adding a new MTR line changes the global Hamiltonian flow and therefore ω at all stations, not just the new ones. Full recompilation of all five passes is required. At 81 stations this takes seconds; at 4B stations it takes hours. The claim that topology extensions are "straightforward" understates this cost.
LLVM analogy is structural, not formal. The Fukaya IR (`au_fukaya_engine.jl`) is a Julia program, not a formally specified intermediate representation. The transformation to flat artifacts cannot be formally verified. The analogy organises the architecture and the thinking; it does not constitute a proof of compilation correctness.
Genuine vs. dressed contributions. We distinguish explicitly:

- *Genuinely theory-derived:* Pass 1 embedding formula; Pass 5 FK brackets; feedback tensor pair projection; dead-zone detection via $W_0(u)\approx 0$; 4ti2 Markov basis computation.
- *Standard engineering in mathematical language:* HNSW hierarchical indexing; EMA feedback updates; m_3 finite-difference stability approximation.

The paper does not always distinguish these clearly enough.

19. Conclusion

We have presented a complete architecture for lazy demographic ad placement grounded in Joyal's Arithmetic Universe and the Fukaya A_∞ category. The three-layer design (AU attic $\to$ Hamiltonian flow $\to$ pair-of-pants Δ) reduces memory from 240 EB to $\approx$ 400 GB, reduces per-slot computation from $O(|S|\cdot|P|)$ to $O(D)+O(\log|P|)$, and produces flat, debuggable, SLA-compliant artifacts through an LLVM-style compilation pass.

The architecture is not merely a clever engineering optimisation. It is a fundamentally different computational model: the precomputed matrix approach requires the closed-world assumption (all combinations known before any decision is made); the AU lazy approach operates under the open-world assumption (objects materialise only when demanded by a surgery node). At planetary scale, only the latter is computationally feasible.

The pharmacodynamic correspondence (Section 11) establishes that the same mathematical structure governs both opioid transport in the mouse brain connectome and product placement in urban transit networks. The 4ti2 computation ($\beta_1 = 37$ MTR Markov circuits, 161 Graver elements) provides the algebraic certificate of the network's toric structure, and the East Rail cross-harbour loop is confirmed as the MTR analogue of the HPF$\leftrightarrow$sAMY dominant oscillator.

Future work: verification of $\partial^2 = 0$ for the MTR Floer complex; deployment of the hierarchical Lagrangian compiler for behavioural cohorts; extension to road networks and the full BALBc 75-node connectome.

References

[1] B. Fang, C.-C. M. Liu, D. Treumann, E. Zaslow, *T-duality and Homological Mirror Symmetry for Toric Varieties*, Adv. Math. **229** (2012), 1873–1911.
[2] K. Fukaya, Y.-G. Oh, H. Ohta, K. Ono, *Lagrangian Floer Theory and Mirror Symmetry on Compact Toric Manifolds*, arXiv:1009.1648v3 [math.SG], 292 pp., 2016.
Exact results used: **Thm 1.3.34** — Kodaira-Spencer ring isomorphism $\mathrm{ks}^b : T_bH(X;\Lambda_0) \xrightarrow{\sim} \mathrm{Jac}(PO^b)$; our routing score is this pairing on the B-side. **Prop 1.3.16** — Jacobian ring splits as $\mathrm{Jac}(PO^b)\otimes\Lambda \cong \prod_{y\in\mathrm{Crit}(PO^b)}\mathrm{Jac}(PO^b;y)$; each active station is one factor; dead zones fail the Morse condition. **§2.3** — Operator $\mathfrak{q}_{\ell;k;\beta}$ counts pseudoholomorphic discs; our m_2 table entries are $\mathfrak{q}_{0;2;\beta}(;L_i,L_j)$ for the dominant disc class. **§2.10** — Seidel homomorphism and quantum Stanley-Reisner ideal SR^ω; our 37-element Markov basis

(4ti2) generates SR^{ω}; each Markov move gives a quantum SR relation; Pass 6 implements these as cluster mutation coefficients at interchange stations. **Thm 1.1.1** — Mirror isomorphism between toric A-model and LG B-model; the Frobenius manifold structure on $H(X;\Lambda_0)$ is the algebraic foundation for the compiler's coherent sheaf output.

[3] K. Fukaya, Y.-G. Oh, H. Ohta, K. Ono, *Lagrangian Floer Theory and Mirror Symmetry on Compact Toric Manifolds*, arXiv:1009.1648v3 [math.SG], Astérisque 376, 2016. Main result: Theorem 1.1.1 — the Kodaira-Spencer isomorphism $\mathrm{ks}_b : H(X;\Lambda_0) \xrightarrow{\sim} \mathrm{Jac}(PO^b)$ is a ring isomorphism between quantum cohomology (A-side) and the Jacobian ring of the LG potential (B-side).

[4] P. Seidel, *The Historical Moran Model*, arXiv:1511.05781 [math.PR], 2015.